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Comparative Analysis of Lipid Profiles Between Controlled and Uncontrolled Type 2 Diabetes Mellitus Patients at Tarakan Regional General Hospital (2022–2023)

ABSTRACT

Background: Type 2 diabetes mellitus (T2DM) is a major contributor to cardiovascular morbidity, largely due to dyslipidemia associated with poor glycemic control. Evidence regarding the relationship between glycemic status and lipid profile alterations in hospital-based populations in Indonesia remains limited. This study aimed to compare lipid profiles between patients with controlled and uncontrolled T2DM and to evaluate their clinical implications.

Methods: This analytic cross-sectional study was conducted at a tertiary care hospital in Jakarta, Indonesia. A total of 120 patients with T2DM were included using consecutive sampling from medical records. Patients were categorized into controlled (HbA1c <7%) and uncontrolled (HbA1c ≥7%) groups. Lipid parameters analyzed included total cholesterol, low-density lipoprotein cholesterol (LDL-C), high-density lipoprotein cholesterol (HDL-C), and triglycerides. Data were analyzed using independent t-test and Mann–Whitney U test, with a significance level of $p<0.05$.

Results: Patients with uncontrolled T2DM demonstrated significantly higher levels of total cholesterol (210.86 ± 51.64 vs. 179.23 ± 43.24 mg/dL; p=0.001), LDL-C (132.62 ± 37.72 vs. 109.70 ± 38.66 mg/dL; p=0.047), and triglycerides [156.00 vs. 131.66 mg/dL; p=0.013] compared with those with controlled diabetes. No significant difference was observed in HDL-C levels (p=0.585). Dyslipidemia was more frequently observed in patients with uncontrolled glycemic status.

Conclusion: Poor glycemic control is significantly associated with a more atherogenic lipid profile in T2DM patients. These findings highlight the importance of optimal glycemic management and routine lipid monitoring to reduce cardiovascular risk.

Keywords: diabetes mellitus, HDL, total cholesterol, LDL, triglycerides

INTRODUCTION

Type 2 diabetes mellitus (T2DM) is a major global health problem and a leading contributor to cardiovascular morbidity and mortality. According to the International Diabetes Federation, Indonesia ranks among the countries with the highest number of individuals living with diabetes, with approximately 19.5 million cases reported in 2021.[1] National data also indicate an increasing prevalence of diabetes, rising from 6.9% in 2013 to 8.5% in 2018.[2] This growing burden highlights diabetes mellitus as a critical public health concern.

T2DM is characterized by insulin resistance and chronic hyperglycemia, both of which contribute to disturbances in lipid metabolism, commonly referred to as diabetic dyslipidemia. This condition is typically marked by elevated triglyceride levels, increased low-density lipoprotein cholesterol (LDL-C), and reduced high-density lipoprotein cholesterol (HDL-C), all of which increase the risk of atherosclerosis and cardiovascular disease.[3] Persistent hyperglycemia further exacerbates oxidative stress and endothelial dysfunction, accelerating vascular damage.[3-5]

Glycemic control, commonly assessed using glycated hemoglobin (HbA1c), plays a crucial role in modulating metabolic and cardiovascular outcomes. Patients with poor glycemic control (HbA1c ≥7%) are more likely to exhibit a more atherogenic lipid profile compared with those with controlled glycemic levels (HbA1c <7%).[4,5] Despite these established mechanisms, evidence examining the association between glycemic status and lipid profile variations in Indonesian hospital-based populations remains limited.

Therefore, this study aimed to compare lipid profiles between patients with controlled and uncontrolled T2DM at Tarakan Regional General Hospital and to evaluate the clinical implications of glycemic status on cardiovascular risk indicators.

METHODS

This analytic cross-sectional study was conducted at a tertiary care hospital in Jakarta, Indonesia, in October 2024. Ethical approval was obtained from the Institutional Ethics Committee (No. 1811/SLKE/TM/UKK W/FKIK/KEPK/TX/2024).

A total of 120 patients with type 2 diabetes mellitus (T2DM) were included using consecutive sampling from medical records. T2DM was defined based on documented glycated hemoglobin (HbA1c) ≥6.5% or fasting plasma glucose ≥126 mg/dL. Patients with incomplete lipid profile data, those receiving lipid-lowering therapy (including statins, fibrates, niacin, or ezetimibe), and those with chronic kidney disease, hyperthyroidism, or liver cirrhosis were excluded.

Patients were categorized into controlled (HbA1c <7%) and uncontrolled (HbA1c ≥7%) groups. The independent variable was glycemic control status, while the dependent variables included lipid parameters: triglycerides, low-density lipoprotein cholesterol (LDL-C), high-density lipoprotein cholesterol (HDL-C), and total cholesterol. Lipid profile measurements were obtained from medical records and were assessed within a clinically relevant timeframe relative to glycemic evaluation.

Data were analyzed using SPSS version 24. Normality of data distribution was assessed prior to analysis. Independent t-test was used for normally distributed variables, while the Mann–Whitney U test was applied for non-normally distributed data. A p-value <0.05 was considered statistically significant.

RESULTS

A total of 120 patients with type 2 diabetes mellitus (T2DM) were included in this study. The variables presented in Table 1 include demographic characteristics (age and sex) and anthropometric measurements, specifically body mass index (BMI). Most participants were in the pre-elderly age group (45–59 years), comprising 62 individuals (52.7%). Females represented the majority of the cohort (n = 73; 60.8%). Based on body mass index (BMI), Obesity Class I (25–29.9 kg/m²) was the most prevalent category (n = 49; 40.8%). Furthermore, uncontrolled diabetes was observed in 77 patients (64.2%), indicating that poor glycemic control predominated in this population.

Table 1. Distribution of Study Participants According to Demographic and Clinical Characteristics

Characteristics	**Frequency (%)**	**Cholesterol Level (mg/dL)**			
		Total Cholesterol	**Triglycerides**	**LDL**	**HDL**
Age(years)					
Adult(19-44)	10 (8,3)	182,5 (105-324)	130 (68-193)	119 (60-225)	42 (12-60)
Pre-Elderly(45-59)	62 (52,7)	**212** (108-323)	**160** (57-828)	**121** (46-211)	46 (23-75)
Elderly (≥60)	48 (40,0)	191 (95-299)	131 (49-366)	106 (19-209)	45 (23-79)
Sex					
Male	47 (39,2)	177 (95-315)	142 (49-828)	110 (19-209)	44 (12-73)
Female	73 (60,8)	**216** (123-343)	143 (57-617)	**129** (45-225)	50 (23-79)
BMI (kg/m²)					
Underweight (<18,5)	7 (5,8)	205 (145-218)	**195** (75-285)	121 (80-144)	43 (26-60)
Normal (18,5-22,9)	14 (11,7)	182 (135-235)	116 (57-213)	101 (45-159)	43 (27-79)
Overweight(23-24,9)	24 (20,0)	186 (108-343)	129 (49-480)	108 (46-209)	45 (23-75)
Obesity Class I (25-29,9)	49 (40,8)	**212** (95-324)	149 (84-828)	**124** (19-225)	47 (12-75)
Obesity Class II (≥30)	26 (21,7)	194 (123-264)	150 (66-385)	117 (75-211)	47 (23-74)
Glycemic Control Status					
Controlled (< 7%)	43 (35,8)	179,23 ±43,24	128 (49-385)	109,70 ±38,66	45,86 ±13,66
Uncontrolled (≥ 7%)	77 (64,2)	**210,86** ±51,64	**156** (57-828)	**122,62** ±37,72	47,29 ±13,68

Analysis of lipid parameters across age groups showed that the pre-elderly group exhibited the highest median concentrations of total cholesterol [212 (108–323) mg/dL], triglycerides [160 (57–828) mg/dL], LDL-C [121 (46–211) mg/dL], and HDL-C [46 (23–75) mg/dL], suggesting a clustering of lipid abnormalities within this age category. When stratified by sex, female patients demonstrated higher median lipid levels compared with males. Median total cholesterol reached 216 (123–343) mg/dL and LDL-C 129 (45–225) mg/dL among females, indicating a less favorable lipid profile in this subgroup.

Across BMI categories, patients classified as Obesity Class I had the highest median total cholesterol and LDL-C levels, measuring 212 (95–324) mg/dL and 124 (19–225) mg/dL, respectively. In contrast, the highest median triglyceride concentration was observed in the underweight group at 195 (75–285) mg/dL, demonstrating variability in lipid distribution across nutritional status. Stratification according to glycemic control revealed that patients with uncontrolled diabetes consistently exhibited higher lipid levels, including total cholesterol of 210 ± 51,64 mg/dL, triglycerides of 156 (57–838) mg/dL, and LDL-C of 122,62 ±37,72 mg/dL.

Table 2. Distribution of Dyslipidemia Among Patients with Type 2 Diabetes Mellitus

Dyslipidemia	**Glycemic Control Status**	
	Controlled (n=43)	Uncontrolled (n=77)
	n (%)	
Dyslipidemia	4 (3,3)	12 (10,0)

As shown in Table 2, dyslipidemia was more prevalent among patients with uncontrolled diabetes, accounting for 12 individuals (10.0%) of the study population.

Table 3. Comparison of Lipid Parameters According to Glycemic Control Status

Lipid Profile	**Glycemic Control Status**		**Nilai p**
	Controlled (n=43)	**Uncontrolled (n=77)**	
LDL	109,70 ±38,66	122,62±37,72	0,047[a]
HDL	45,86±13,66	47,29±13,68	0,585[a]
Kolesterol Total	179,23±43,24	210,86±51,64	0,001[a]
Trigliserida	131,66 (49-285)	156,00 (57-828)	0,013[b]

[a]Uji T test independent, [b]Uji mann whitney, significant if $p<0,05$

The results demonstrated significant differences in total cholesterol ($p = 0.001$), LDL-C ($p = 0.047$), and triglyceride levels ($p = 0.013$) between patients with controlled and uncontrolled type 2 diabetes mellitus. Overall, patients with uncontrolled diabetes exhibited higher lipid levels compared with those with controlled diabetes. The mean LDL-C level was 122.62 ± 37.72 mg/dL, and the mean total cholesterol level was 210.86 ± 51.64 mg/dL. Meanwhile, the median triglyceride level was 156.00 (57–828) mg/dL.

DISCUSSION

This study demonstrated that most patients with type 2 diabetes mellitus (T2DM) were in the pre-elderly age group (45–59 years), which also showed higher lipid levels compared to other age groups. These findings are consistent with previous studies reporting that individuals aged above 45 years are more susceptible to metabolic disturbances, including dyslipidemia, due to age-related physiological changes and unhealthy lifestyle patterns.[9,10] Age-related decline in metabolic function, particularly impaired hepatic and pancreatic activity, may contribute to both glucose and lipid dysregulation.[10,11]

Female patients constituted the majority of the study population and exhibited higher lipid levels compared to males. Similar findings have been reported in previous studies, where women with T2DM showed a higher prevalence of dyslipidemia.[7,12] Hormonal factors, particularly decreased estrogen levels, may contribute to increased visceral fat accumulation, insulin resistance, and enhanced hepatic lipoprotein production, thereby influencing lipid metabolism.[11]

In terms of nutritional status, most patients were classified as Obesity Class I, which was associated with higher total cholesterol and LDL levels. These findings are in line with previous studies demonstrating that obesity is strongly associated with lipid abnormalities and insulin resistance.[13],[14] Excess adipose tissue contributes to increased release of free fatty acids and pro-inflammatory cytokines, which disrupt insulin signaling and promote hepatic lipoprotein synthesis, thereby exacerbating dyslipidemia.[13],[15]

Interestingly, lipid abnormalities were also observed in underweight individuals, particularly elevated triglyceride levels. Although evidence is limited, this finding suggests that dyslipidemia in T2DM is not solely associated with obesity but may also be influenced by metabolic stress and underlying pathological conditions.[14],[16] This highlights the complex nature of lipid metabolism disturbances in diabetic patients.

The present study demonstrated that patients with uncontrolled T2DM had significantly higher levels of LDL-C, total cholesterol, and triglycerides compared with those with controlled diabetes. These findings are consistent with previous studies showing that poor glycemic control is associated with a more atherogenic lipid profile.[4],[5],[15] Elevated LDL-C levels in uncontrolled diabetes may be attributed to increased hepatic very low-density lipoprotein (VLDL) production and subsequent formation of atherogenic lipoprotein particles.[18],[19]

In contrast, HDL-C levels did not differ significantly between groups, although slightly higher levels were observed in patients with uncontrolled diabetes. Previous studies generally report reduced HDL-C levels in T2DM due to increased catabolism of HDL particles and impaired reverse cholesterol transport.[19] The lack of significant difference in this study may be influenced by variations in patient characteristics, treatment adherence, or sample size.

Higher triglyceride levels observed in patients with poor glycemic control further support the role of insulin resistance in lipid metabolism disorders. Insulin resistance promotes increased lipolysis and free fatty acid release, leading to enhanced hepatic triglyceride synthesis and VLDL production.[19] These mechanisms contribute to the development of diabetic dyslipidemia and increased cardiovascular risk.

Overall, this study confirms that poor glycemic control plays a central role in the development of dyslipidemia in T2DM. Patients with uncontrolled diabetes exhibit a more atherogenic lipid profile, characterized by elevated LDL-C, total cholesterol, and triglyceride levels. These findings emphasize the importance of optimal glycemic control and routine lipid monitoring as part of comprehensive diabetes management to reduce cardiovascular risk.

CONCLUSION

This study demonstrates that glycemic status plays a critical role in shaping lipid abnormalities in patients with type 2 diabetes mellitus. Uncontrolled diabetes was consistently associated with significantly higher levels of total cholesterol, LDL-C, and triglycerides, reflecting a more atherogenic lipid profile and a greater potential for cardiovascular complications. Notably, lipid disturbances were observed across different BMI categories, including underweight individuals, indicating that dyslipidemia in T2DM is not solely attributable to obesity but arises from complex metabolic interactions involving insulin resistance, glycemic control, and lipid metabolism. These findings underscore the importance of comprehensive metabolic assessment and strict glycemic management, alongside routine lipid monitoring, to reduce cardiometabolic risk and improve long-term outcomes in patients with T2DM.

Conflict of Interest
The authors declare that there are no conflicts of interest regarding the publication of this study.

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